

Pathway enrichment shifts from H2BK12ac at 6hr to H3K27ac at 24hr

Pathway-level enrichment was assessed for H2BK12ac, H3K27ac, and H4K5ac using CRF pair-defined narrowPeak sets derived from mm10-aligned tracks at 6 hr and 24 hr. At 6 hr, enrichment was strongly polarized toward H2BK12ac, which preferentially marked interferon- and inflammation-associated programs (INTERFERON_ALPHA_RESPONSE, INTERFERON_GAMMA_RESPONSE, IL6_JAK_STAT3_SIGNALING, INFLAMMATORY_RESPONSE, TNFA_SIGNALING_VIA_NFKB), together with stress/apoptosis-related sets (UV_RESPONSE_UP, APOPTOSIS, HYPOXIA) (Figure 1). In contrast, H3K27ac and H4K5ac showed sparse or weak associations across these pathways at 6 hr, and metabolic (e.g., CHOLESTEROL_HOMEOSTASIS, FATTY_ACID_METABOLISM) and proliferation/cell-cycle signatures (E2F_TARGETS, G2M_CHECKPOINT) were notably depleted across marks (Figure 1).

By 24 hr, the enrichment landscape shifted to H3K27ac dominance, with robust signal across immune and cytokine-related pathways (IL6_JAK_STAT3_SIGNALING, IL2_STAT5_SIGNALING, INTERFERON_GAMMA_RESPONSE, TNFA_SIGNALING_VIA_NFKB) (Figure 2). INTERFERON_ALPHA_RESPONSE and INFLAMMATORY_RESPONSE were comparatively weaker than at 6 hr, and most other categories—including MYC_TARGETS, OXIDATIVE_PHOSPHORYLATION, and EPITHELIAL_MESENCHYMAL_TRANSITION—showed little to no enrichment. H2BK12ac and H4K5ac remained broadly weak at 24 hr (Figure 2).

Together, these results indicate a temporal shift in immune-pathway engagement from H2BK12ac-biased interferon/inflammatory programs at 6 hr to an H3K27ac-dominant

immune-signaling profile by 24 hr, with consistently sparse enrichment of metabolic and proliferation/cell-cycle pathways across marks and time points.

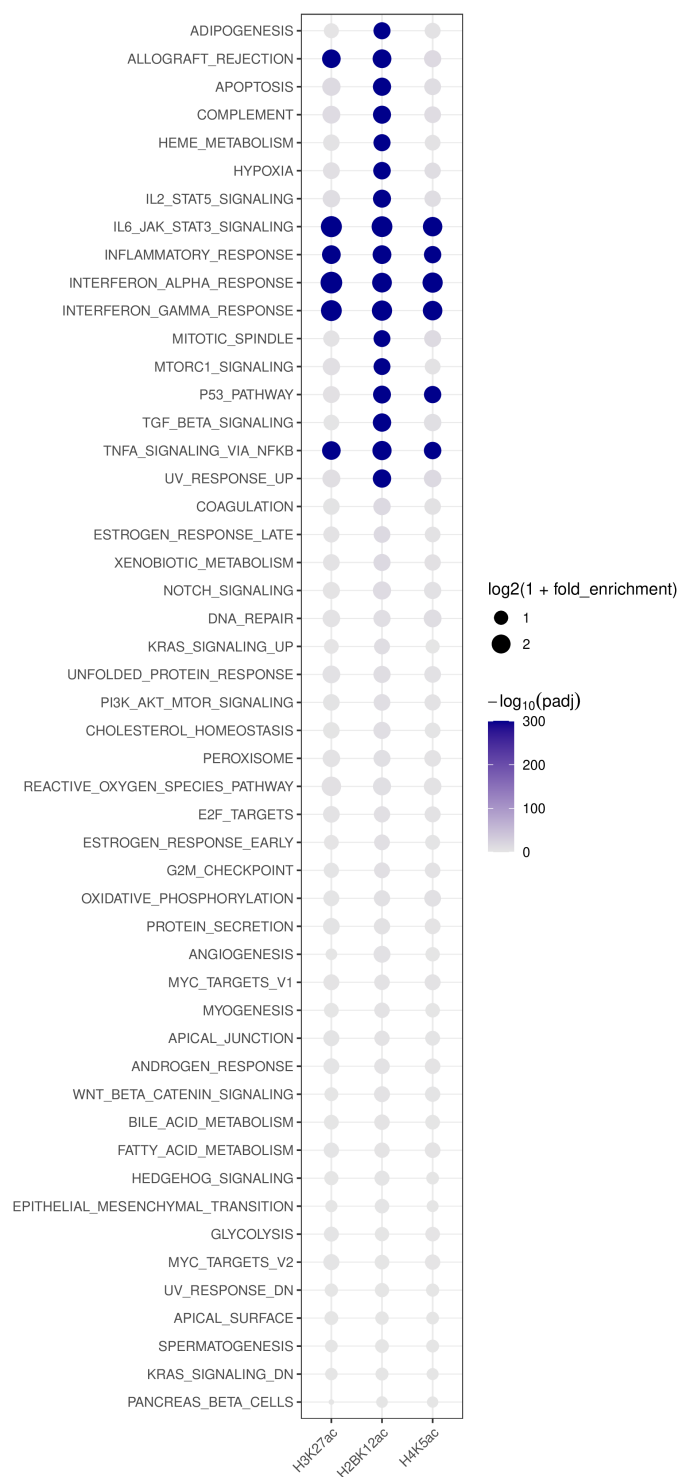


Figure 1. Pathway enrichment concentrates in H2BK12ac at 6hr. Hallmark gene-set enrichment of CRF pair-specific peak sets from the 6hr sample is shown for H3K27ac, H2BK12ac, and H4K5ac. The dominant signal is confined to H2BK12ac, highlighting interferon, inflammatory, TNFA/IL6–STAT3, apoptosis, hypoxia, and UV-response pathways, while H3K27ac and H4K5ac show minimal enrichment across most categories.

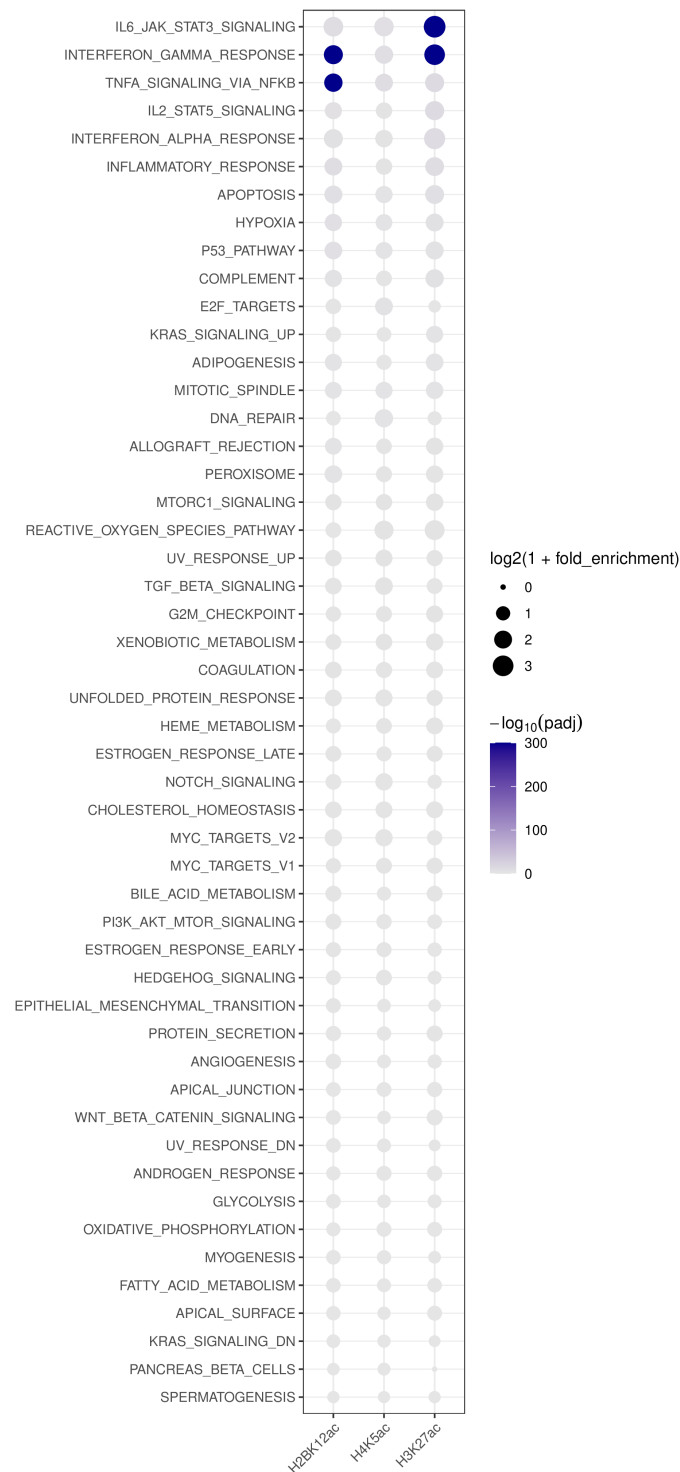


Figure 2. 24hr H3k27ac peaks enrich immune signaling pathways. Pathway enrichment of 24hr peak sets across three histone marks highlights selective enrichment in H3k27ac, with prominent signals for *IL6_JAK_STAT3_SIGNALING*, *IL2_STAT5_SIGNALING*, *INTERFERON_GAMMA_RESPONSE*, and *TNFA_SIGNALING_VIA_NFKB*. H2bK12ac and H4K5ac show minimal enrichment across most pathways.

Candidate regulators from TF motif enrichment at 6hr and 24hr

Motif enrichment analyses across chromatin marks and time points nominate a coherent set of candidate regulators with mark- and stage-selective patterns. At 6 hr, H3K27ac peaks display the strongest and broadest motif over-representation (Figure 3), with prominent signals for ETS family (e.g., ETS1), IRF (IRF4), MYC/MAX, and EP300 motifs, alongside POLR2A. Given that p300/CBP catalyzes H3K27 acetylation at active enhancers (Araki & Mimura, 2017; Wan et al., 2022), and ETS factors can recruit p300 to drive H3K27ac accumulation at limb enhancers (Peluso et al., 2017), these enrichments collectively point to an early ETS–IRF–MYC axis operating on H3K27ac regions, with EP300 as a likely coactivator. In contrast, JUN/AP-1 shows a relative preference for H2BK12ac at 6 hr (Figure 3), nominating AP-1 as an early regulator of this mark, whereas CTCF, E2F4, NANOG, and PAX5 remain near baseline and are unlikely primary drivers at this stage.

By 24 hr, H3K27ac continues to concentrate the strongest motif signals across factors (Figure 4). ETS, IRF4, JUN, MAX, and SIN3A remain enriched, but several features distinguish the late state: GABPA exhibits a pronounced H3K27ac-specific motif peak (Figure 4), consistent with its ETS-family role and the capacity to recruit p300 to activate distal enhancers (Peluso et al., 2017), nominating GABPA as a key 24 hr regulator on H3K27ac sites. In parallel, POLR2A (Ser5P)-associated motifs are selectively enriched in H3K27ac and H4K5ac but remain near baseline in H2BK12ac (Figure 4), aligning with the association of RNAPII Ser5P and general transcription machinery at active enhancers and promoters (Wan et al., 2022). H2BK12ac retains an overall lower motif load, with JUN remaining comparatively stronger than many other factors, whereas H4K5ac displays intermediate enrichment patterns.

Integrating marks and time points, we nominate: - Early (6 hr) H3K27ac-driven regulators: ETS (e.g., ETS1) and IRFs (e.g., IRF4), together with MYC/MAX, with EP300 as a coactivator consistent with H3K27ac deposition (Araki & Mimura, 2017; Wan et al., 2022). - Early (6 hr) H2BK12ac-biased regulator: JUN/AP-1. - Late (24 hr) H3K27ac regulators: GABPA (ETS family), with p300 cooperation (Peluso et al., 2017), accompanied by sustained ETS/IRF/MYC/MAX signals. - Late (24 hr) transcriptional engagement: POLR2A (Ser5P) enrichment on H3K27ac/H4K5ac peak sets, indicating increased RNAPII Ser5P occupancy at sustained regulatory elements (Wan et al., 2022).

Collectively, the data support a model in which an early ETS–IRF–MYC program establishes enhancer activity on H3K27ac, AP-1 transiently marks H2BK12ac sites at 6 hr, and by 24 hr, GABPA–p300 cooperation consolidates H3K27ac enhancers with increased

RNAPII Ser5P engagement. These candidates provide mechanistic anchors for downstream validation.

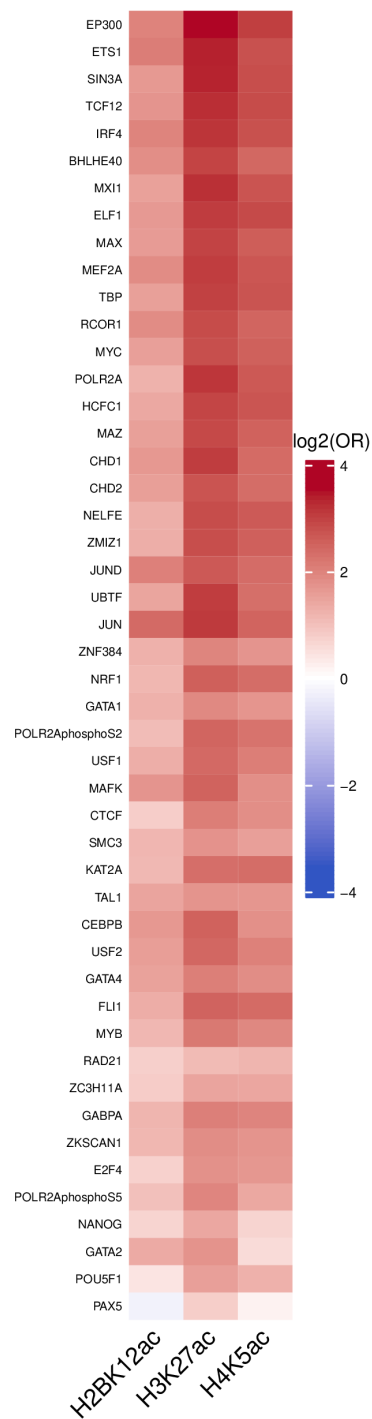


Figure 3. TF motif enrichment across histone-marked peak sets at 6 hr. Heatmap of $\log_2$ odds ratios for TFBS motifs within peak sets defined by H2BK12ac, H3K27ac, and H4K5ac. Enrichment is broadly positive, with H3K27ac peaks showing the highest motif enrichment for many factors, whereas CTCF, E2F4, NANOG, POU5F1, and PAX5 are relatively weak.

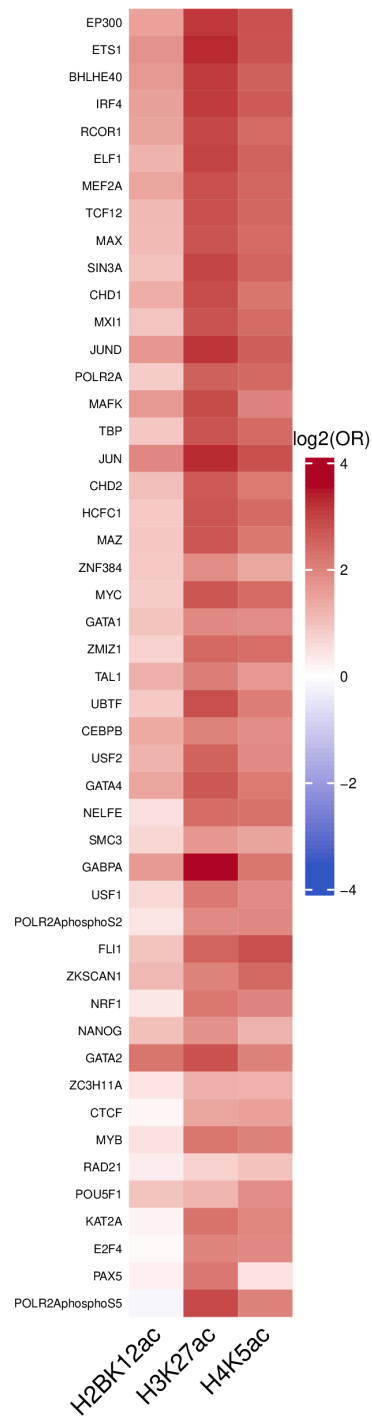


Figure 4. TF motif enrichment across 24hr acetylation peak sets. Heatmap of $\log_2$ odds ratios for motif over-representation in 24hr peaks marked by H2BK12ac, H3K27ac, and H4K5ac. Enrichment is broadly positive, with H3K27ac yielding the strongest signals for many TFs; GABPA and JUN are high, whereas CTCF, RAD21, and E2F4 are weaker.

Regulatory element classes and gene-context localization across time

Intersecting peaks with ENCODE cCREs and gene models revealed a consistent, enhancer-biased regulatory architecture at both 6 hr and 24 hr that localizes predominantly within intronic space. At 6 hr, dELS and pELS constituted the largest cCRE fractions for all marks, with a smaller but detectable PLS component and minimal TF/CTCF-associated categories (Figure 5). Within this shared enhancer bias, H2BK12ac (and to a lesser extent H4K5ac) showed an expanded “other” category and the most apparent CA (CTCF-only) contribution, whereas H3K27ac displayed the smallest “other” share. The 24 hr cCRE profiles recapitulated these patterns: H3K27ac retained the strongest enhancer/promoter-like representation (dELS/pELS with a clear PLS), H4K5ac was similar but more “other”-enriched, and H2BK12ac exhibited the smallest pELS/PLS and the largest “other” fraction while maintaining a notable dELS component (Figure 7).

Gene-centric localization paralleled the cCRE assignments. At 6 hr, peaks were predominantly intronic, with promoters as the next most frequent category; exons and UTRs were minor across marks (Figure 6). H2BK12ac showed a smaller promoter share and the largest “other” assignment relative to H3K27ac and H4K5ac. At 24 hr, introns again dominated across all marks, with promoters secondary and exons/UTRs remaining modest; H2BK12ac displayed lower exonic representation and a slightly larger “other” component than the other marks (Figure 8).

Together, these distributions define a stable, enhancer-centric, intron-dominant regulatory landscape from 6 hr to 24 hr. Against this backdrop, H2BK12ac is distinguished by reduced

promoter-like contributions and elevated “other”/CA components compared with H3K27ac, with H4K5ac generally intermediate (Figures 5–8).

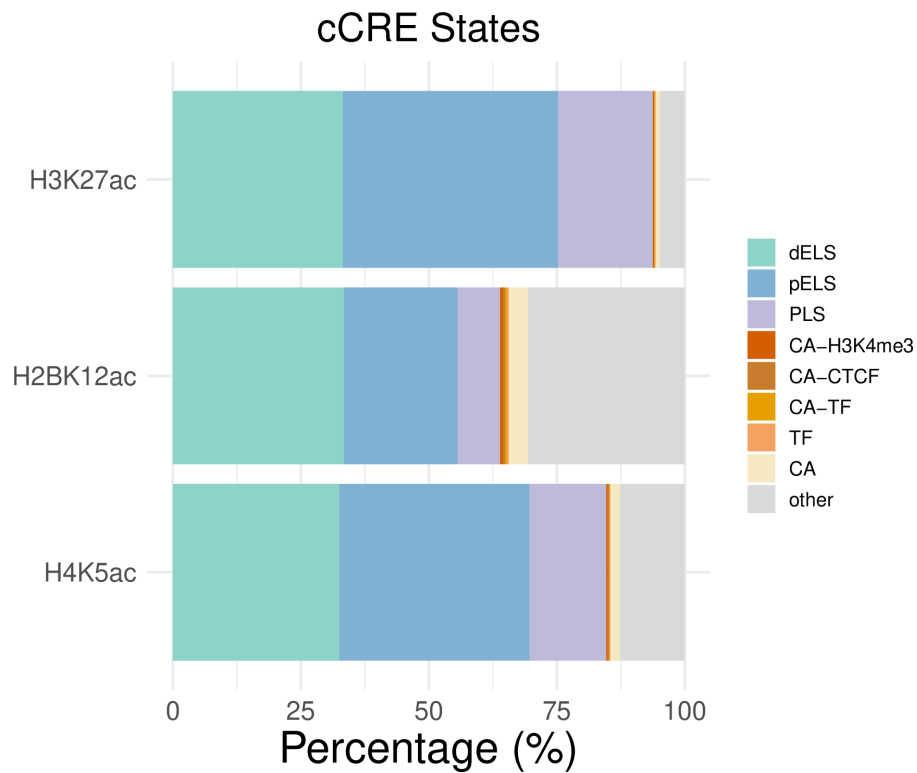


Figure 5. 6hr peak assignments across ENCODE cCRE categories. 6hr peaks for H3K27ac, H2BK12ac, and H4K5ac are assigned to ENCODE cCRE states. Enhancer-like classes (dELS, pELS) dominate across marks, with PLS secondary; TF/CTCF-associated and CA subclasses are minor, and the “other” fraction is larger for H2BK12ac and H4K5ac than for H3K27ac.

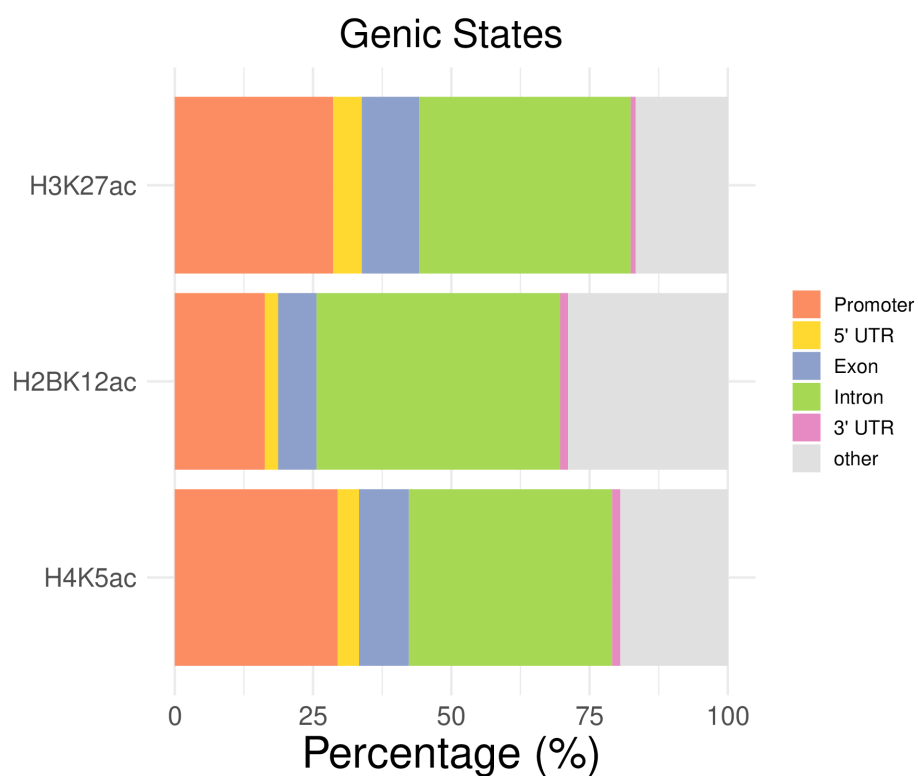


Figure 6. Genic distribution of 6hr peaks by histone mark. Peaks mapped to mm10 gene annotations are summarized for H3K27ac, H2BK12ac, and H4K5ac. Intronic assignments dominate for all marks, with promoters contributing the next-largest share for H3K27ac and H4K5ac; exons and UTRs are minor. The other category is comparatively higher for H2BK12ac.

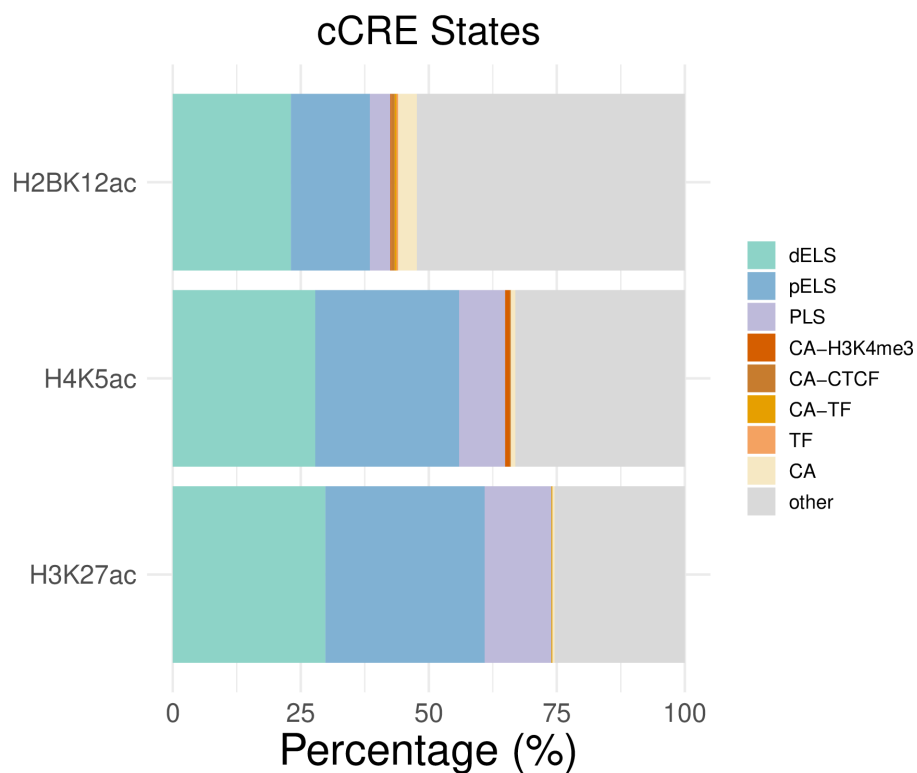


Figure 7. cCRE class composition of 24hr peak overlaps. Percentages of 24hr peaks overlapping ENCODE cCRE classes for H2BK12ac, H4K5ac, and H3K27ac. dELS and pELS account for most overlaps in H3K27ac and H4K5ac, whereas H2BK12ac shows a larger “other” fraction and fewer PLS/pELS; CA-associated and TF categories are minor.

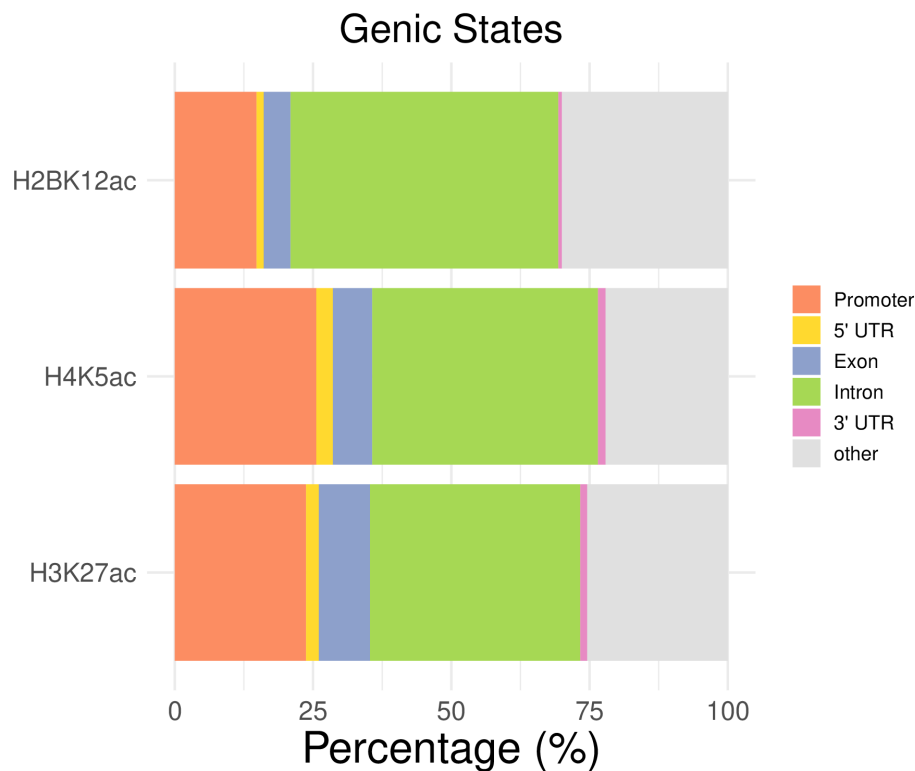


Figure 8. Genic distribution of 24hr peaks for three acetylation marks. Proportions of 24hr peaks assigned to promoters, exons, introns, UTRs, and other regions are shown for H2BK12ac, H4K5ac, and H3K27ac. All three marks are dominated by intronic assignments with a substantial promoter component, while exonic and UTR fractions are small. H2BK12ac displays comparatively lower promoter and exon shares than the other two marks.

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